

YUANKAI TAO

Email: yuankaitao0909@gmail.com | Tel: +1 470 658 3699 | [GitHub](#) | [Homepage](#)

EDUCATION

Georgia Institute of Technology <i>M.S. in Operations Research, Stewart School of Industrial and Systems Engineering</i> Selected Coursework: Optimization, Stochastic Processes, Simulation	<i>Aug 2026 — Dec 2027</i>
University of California, San Diego <i>M.S. in Quantitative Finance, Rady School of Management</i> Selected Coursework: Data Science for Finance, Deep Learning, Computational Finance Methods	<i>Sep 2024 — Dec 2025</i> GPA: 3.84/4.0
Zhongnan University of Economics and Law <i>B.S. in Management, School of Accounting and Finance</i> Selected Coursework: Probability & Statistics, Linear Algebra, Calculus, Python Programming	<i>Sep 2020 — Jun 2024</i> GPA: 3.63/4.0

TECHNICAL SKILLS

Programming & Querying	Python, SQL, MATLAB, STATA, VBA, $\text{\LaTeX}$
Data & Cloud	PostgreSQL, PostGIS, BigQuery, DuckDB, GCP, Linux, Git
ML & Analytics	Statistical Modeling, Time Series, Feature Engineering, Model Selection, Validation
Libraries & Tools	Pandas, Polars, NumPy, scikit-learn, PyTorch, Matplotlib, LightGBM, XGBoost

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill <i>The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining</i>	<i>Mar 2025 — Jan 2026</i>
- Engineered an end-to-end Python and SQL pipeline for a 76-million-observation spatial dataset with 40+ variables on Google Cloud Platform, integrating property and population data from multiple sources. - Developed parallelized PostGIS workflows with property-specific filters across multiple distance ranges, achieving approximately 50$\times$ computational speedup while preserving reproducible analytical logic. - Built and managed a 964.5M-record, 100GB+ database using PostgreSQL, BigQuery, and DuckDB on GCP and UNC Longleaf HPC; executed complex geospatial and time-window queries for downstream research.	

PROFESSIONAL EXPERIENCE

VariFlight Technology Data Analyst, Intern	<i>Jul 2025 — Sep 2025</i> <i>Data Development Department Anhui, China</i>
- Built Python web-scraping workflows to collect public reports, webpages, and images from the Civil Aviation Administration of China for recurring data analysis. - Applied OCR and table-extraction tools to transform 600+ unstructured PDF reports and 500+ web-pages into standardized, analysis-ready datasets for local LLM training; added repeatable cleaning and quality-control steps. - Created dashboards for transportation and passenger-turnover metrics, visualizing operational trends and enabling faster interpretation of key aviation indicators.	
Guangdong Hedge Fund Investment Management Co., Ltd. Quantitative Research, Intern	<i>May 2026 — Jul 2026</i> <i>Quantitative Strategy Department Hong Kong</i>
- Developed an end-to-end ML pipeline with 16 trend, momentum, volatility, liquidity, volume, and open-interest features; trained LightGBM, XGBoost, CatBoost, and neural-network models across 1- to 20-day horizons. - Designed rolling, time-aware validation and retraining workflows; evaluated models with statistical, risk, and transaction-cost metrics to reduce leakage and support reproducible comparison. - Implemented a monthly rebalanced model in Backtrader that achieved a 1.89 Sharpe ratio, 16.79% annualized return, and 25.51% excess return after transaction costs.	

Ernst & Young

Audit, Intern

Dec 2022 — Feb 2023

Financial Service Office (FSO) | Beijing, China

- Automated the cleaning and consolidation of financial tables from **220+ CSV files**, generating standardized audit outputs and improving the consistency of recurring data-review workflows.
- Performed variance analysis on intercompany and investment transactions, investigated anomalies, and validated findings against supporting records.

SELECTED PROJECTS

Kaggle Jane Street Market Prediction Competition

Silver Medal

- Processed **47M minute-level observations** spanning 1,669 trading days; engineered cross-sectional, lagged, rolling, and regime features from 79 anonymized variables and 9 targets.
- Trained and ensembled LightGBM, CatBoost, XGBoost, MLP, and TabNet models with time-aware validation; placed **94th (Top 2.5%)** of 3,757 teams on the private leaderboard.

Text Analysis and BERT Fine-tuning on SEC 10-K Reports

MGTF 423, UCSD

Team Lead | Analysis and Human-in-the-Loop Learning on 20.5M Text Rows

- Performed named-entity recognition, frequency analysis, and visualization on SEC filings to identify patterns in corporate disclosures.
- Built a human-in-the-loop labeling and fine-tuning workflow with three annotation rounds, achieving **79% accuracy** in classifying uncertainty tone under limited labeled data.

UCSD / FMSbonds, Inc. 2025 Capstone Project

Aug 2025 — Dec 2025

Team Lead | High-Performance Bond Pricing Engine Optimization

- Designed caching for high-frequency cash-flow queries, eliminating redundant object generation and reducing per-bond runtime from 1.17 ms to 0.13 ms (**9× speedup**).
- Reimplemented the yield-to-worst solver with Numba JIT, reducing runtime from 1.57 ms to 7.5 μ s per bond (**210× speedup**) for scalable batch analytics.

CERTIFICATIONS & HONORS

Deep Learning Specialization (5-Course Certificate)

Coursera, DeepLearning.AI

Algorithms and Data Structures Certificate (6-Course Certificate)

Coursera, UC San Diego

Beta Gamma Sigma (Top 20% of graduating class)

UCSD, 2024 — 2025

WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%)

2024

Special Scholarship in Innovation and Entrepreneurship

ZUEL, 2022 — 2023